HUBO Degree Reduction for Pangenome-Guided Assembly

Concise Results Summary

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Purpose and scope

The reported experiments reduce eighth-order higher-order unconstrained binary optimisation (HUBO) models to maximum degree six (D6) or five (D5). Three graph methods—Minigraph, Minigraph-Cactus, and PGGB—are compared at 12, 16, and 20 original binary variables, giving 18 configurations. The results below are transcribed from the supplied presentation (slides 2–7); they are not a new execution of the pipeline. The background paper motivates the broader trade-off between compact HUBO encodings and quantum circuit complexity [1].

Computational results

Reduction decreases the largest number of variables interacting in a term, but introduces auxiliary binary variables. The same variable counts are reported for all three graph methods.

Original variables	Degree change	Auxiliaries	Total variables	Overhead	Space multiplier
12	$8 \to 6$	3	15	25.0%	8×
12	$8 \to 5$	5	17	41.7%	32×
16	$8 \to 6$	4	20	25.0%	16×
16	$8 \to 5$	7	23	43.8%	128×
20	$8 \to 6$	5	25	25.0%	32×
20	$8 \to 5$	9	29	45.0%	512×

Scaling explanation. For q original variables and a auxiliaries, the number of possible binary assignments increases from 2^q to 2^{q+a}, a factor of 2^a. These multipliers are calculated from the reported variable counts; they are not measured runtime increases. Full statevector storage also scales as 2^{q+a}, although other solvers need not store every state. At 20 original variables, D5 has $2^{29-25} = 16$ times as many assignments as D6. D6 therefore has lower variable overhead throughout the tested sizes.

Main biological findings

Fifteen of the 18 configurations have no reported change in identity, mismatches, insertions, or deletions. The three exceptions improve identity:

- **Minigraph, 16 variables, D5:** identity increases by 4.64 percentage points, with 33 fewer mismatches and two fewer deletions, but one additional insertion.
- **PGGB and Minigraph-Cactus, 16 variables, D6:** each improves identity by 0.25 percentage points, with two fewer mismatches and no insertion/deletion change.

No displayed configuration loses identity, but neither degree consistently improves it. These results concern changes in the reported sequence metrics; unchanged metrics alone do not prove that the decoded sequence or graph walk is identical.

Simple explanation. Think of a genome as an essay assembled from linked sentences. Reduction simplifies the rules connecting the sentences, but adds extra bookkeeping choices. Most reported essays retain the same accuracy; three improve. More bookkeeping can make searching for the best essay harder.

Complete biological comparison

All changes are reduced minus unreduced. Identity is in percentage points (pp); negative mismatch, insertion, or deletion changes mean fewer reported errors. The presentation does not specify whether insertion/deletion counts represent events or affected bases, so that unit is left unspecified here. Original denotes the original binary-variable count, labelled Qubits in the slides; it is not evidence of a hardware run.

Method	Target	Original	Δ Identity (pp)	Δ Mismatch	Δ Insertion	Δ Deletion
Minigraph	D6	12	0.00	0	0	0
Minigraph	D5	12	0.00	0	0	0
Minigraph	D6	16	0.00	0	0	0
Minigraph	D5	16	+4.64	−33	+1	−2
Minigraph	D6	20	0.00	0	0	0
Minigraph	D5	20	0.00	0	0	0
Minigraph-Cactus	D6	12	0.00	0	0	0
Minigraph-Cactus	D5	12	0.00	0	0	0
Minigraph-Cactus	D6	16	+0.25	−2	0	0
Minigraph-Cactus	D5	16	0.00	0	0	0
Minigraph-Cactus	D6	20	0.00	0	0	0
Minigraph-Cactus	D5	20	0.00	0	0	0
PGGB	D6	12	0.00	0	0	0
PGGB	D5	12	0.00	0	0	0
PGGB	D6	16	+0.25	−2	0	0
PGGB	D5	16	0.00	0	0	0
PGGB	D6	20	0.00	0	0	0
PGGB	D5	20	0.00	0	0	0

Interpretation and next steps

The displayed results support preservation of the measured assembly accuracy in most configurations, with occasional improvement. D6 uses fewer auxiliaries; D5 offers a lower maximum interaction degree but a larger search space. These tables alone do not establish which degree is faster, requires shallower circuits, or performs better on quantum hardware. Improvement in sampled solutions also does not, by itself, prove mathematical equivalence of the reduced and original objectives.

The supplied comparison lacks absolute identity values, coverage, assembly break counts, timings, circuit statistics, and repeated-run uncertainty. Accordingly, it cannot establish complete genome reconstruction or a statistically reliable advantage. The next useful checks are to validate legal graph walks and auxiliary consistency, compare decoded sequences against ground truth, and measure runtime and circuit resources across repeated runs before assessing IonQ hardware performance.

References

[1] J. Cudby and S. Strelchuk, *Nonvariational quantum optimisation approaches to pangenome-guided sequence assembly*, arXiv:2604.06106v1, 2026. Supplied PDF: *NonVariational .pdf*. `https://arxiv.org/abs/2604.06106`